

Datathon #4 - Neural Networks to Predict Physical Activity

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Introduction: A recent World Health Organization report found that almost 1.8 billion adults worldwide did not meet recommended physical activity benchmarks in 2022.¹ Physical activity-related health promotion strategies have lagged behind the available evidence; mobile Health (mHealth) applications represent an emerging technology for health promotion.² Machine learning (ML)-supported mHealth has the transformative potential to enable proactive health monitoring, bolster health management, and promote good health.

In this study, we harnessed sensor data to deploy a Long Short-Term Memory (LSTM) neural network that predicts the activity levels of individuals. This analysis provides valuable insights into how ML-supported mHealth applications can identify individuals below recommended physical activity thresholds and tailor interventions accordingly to promote active lifestyles.

Dataset Description: This dataset was sourced from Sunnybrook Health Sciences Center and contains sequential time-series sensor data from 9 healthy participants (median age = 22.5; median BMI = 24). Participants wore a sensor on their left ankles and right arms, and gyroscopic and acceleration measurements were collected at 50 Hz. These measurements were annotated with their corresponding activity levels. There were 13 total activity levels (our classification outcome classes), ranging from 0 ('Nothing') to 12 ('Jumping front and back'). Classes 0 - 3 correspond to sedentary activities, classes 4 -8 corresponded to light activity/stretching, and classes 9-12 corresponded to vigorous activity.

Data Engineering: We created a filtered dataset that grouped items by 'subject.' Descriptive statistics were calculated for the remaining columns, including column means, standard deviations, minimums, and maximums. Data missingness was appraised, however, we had no data missingness. Time domain features, including Root Mean Square (RMS), Signal Magnitude Area (SMA), Jerk (rate of change of acceleration), and motion variance were computed. Using the timestamp column, we confirmed that the dataset was continuous. Sliding window segmentation was performed to convert raw time-series data to group readings into motion patterns. Our sampling rate was set at 50 (50 samples per second), our window size was set at 2 (2 second windows), and our stride was set to a 50% overlap. Our feature matrix shape was (19998, 132).

Model Justification: We deployed an LSTM neural network model to predict activity classifications across sequential data over time. An LSTM neural network was selected because of its ability to retain pertinent context over time. This enabled the model to make predictions based on historical, sequential patterns. More broadly, a neural network architecture was chosen because of its ability to model non-linear patterns within the data and perform multi-class classification.

Analysis: Before deploying our model, we normalized the features, performed one-hot encoding on the labels, reshaped the input data, and split the data into training and testing sets (80:20). The model contained 64 neurons to empower more complex pattern recognition. Further, we specified a neuron dropout rate of 0.3 during training to prevent overfitting. This process was repeated during the second layer of the LSTM. The final layer contained neurons equal to the number of classes. Softmax activation was specified to ensure classification. When compiling the model, the ADAM optimizer was deployed to automatically improve training efficiency. The categorical cross-entropy loss function enabled multi-class classification, and accuracy metrics were collected. The trained model reviewed 20 epochs and had a batch size of 32. Figures depicting model loss (training versus validation loss across epochs) and model accuracy (training versus validation accuracy across epochs) were plotted. Furthermore, a confusion matrix was created to visualize model performance across cases. Finally, a classification report was generated to evaluate model performance.

Findings: The classification report revealed that our model performed strongly across classes. Overall, the model demonstrated an accuracy of 0.91 across 4000 cases. F1 scores varied across classes but ranged from 0.75 (class 1) to 0.94

(class 0). Recall scores ranged from 0.69 (class 8) to 0.99 (class 10, class 11). Precision scores ranged from 0.69 (class 9) to 0.97 (class 1). F1 scores are a particularly relevant metric when assessing the efficacy of our model because it accounts for over-inflated accuracy as a result of class imbalance. The full classification report is provided below (Figure 1). Our model generally performed better in the classification of no activity and vigorous activity classes than light activity. It was interesting to see our model perform better in vigorous activity class classification than moderate activity, especially when both groups constituted the minority class.

Figure 1. Classification Report

There are three important limitations of this model that may its clinical relevance. First, without supplemental analyses, the box’ nature makes it difficult to discern the features that most significantly contributed to the model’s predictive prowess. In practice, the lack of interpretability makes it difficult to draw conclusions about potential relationships that may exist with data. Second, class imbalance within our dataset means that overall, our model showed better performance when classifying cases in the majority class compared to the minority classes. This discrepancy would be best rectified with class distribution. Finally, this model operates on data derived young, healthy individuals. As such, the sample demographics necessarily reflect the wider population. Consequently, the model’s performance may falter when introducing more diverse participants into the dataset or testing against more diverse

	precision	recall	f1-score	support	limit
0	0.97	0.91	0.94	2912	‘black
1	0.72	0.79	0.75	102	
2	0.76	0.94	0.84	103	
3	0.75	1.00	0.86	94	firm
4	0.86	0.93	0.89	81	
5	0.83	0.85	0.84	100	
6	0.72	0.88	0.79	81	the
7	0.79	0.95	0.86	106	
8	0.69	0.95	0.80	81	
9	0.90	0.96	0.93	99	better
10	0.88	0.99	0.93	100	
11	0.86	0.99	0.92	102	
12	0.71	0.90	0.80	39	from 9
accuracy			0.91	4000	
macro avg	0.80	0.93	0.86	4000	
weighted avg	0.92	0.91	0.92	4000	do not

Conclusion: In conclusion, our LSTM neural network demonstrates better than chance performance when classifying cases across activity classes. However, limitations related to interpretability, class imbalance, and dataset diversity reduce the model’s clinical relevance and ability to inform mHealth applications. At this stage, we recommend further testing to confirm the model’s performance across diverse datasets. Future work may include seeking out supplemental data sources to expand the dataset’s class representation and participant diversity, and evaluating the model further to assess overfit. Moreover, future work may seek to embed a SHAP analysis to enhance the model’s interpretability. In doing so, a potential mHealth application leveraging this model is better equipped to make meaningful predictions about activity to drive better health outcomes.

Contributions: The three team members independently performed exploratory analyses of the dataset in Python. Jenny created the final model. Batool compiled the written report. Marya compiled the presentation slide deck. All three members communicated together and collaboratively worked to edit, amend, and contextualize the code, iterate upon the model, and finalize the written report and slides.

Link to Code: https://colab.research.google.com/drive/12Vc2n3I_eyVZVyJwki5tNNM9MEvLdTas?usp=sharing
Presentation: https://docs.google.com/presentation/d/107FZRbgczt1sM8pOD75P3AONGreLzNWKfz0SMvsnxtI/edit?usp=drive_link
Github Repo: <https://github.com/Marya1ZZ/UofT/tree/Dataathon-4>
References

1. Nearly 1.8 billion adults at risk of disease from not doing enough physical activity [Internet]. [cited 2025 Mar 11]. Available from: <https://www.who.int/news/item/26-06-2024-nearly-1.8-billion-adults-at-risk-of-disease-from-not-doing-enough-physical-activity>
2. Kohl HW, Craig CL, Lambert EV, Inoue S, Alkandari JR, Leetongin G, et al. The pandemic of physical inactivity: global action for public health. Lancet. 2012 Jul 21;380(9838):294–305.